

Prime-Side Completion Addendum

Proved estimates, integration criterion, and the remaining comparison

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1. What the supplied work establishes

This addendum develops the proposed completion in number-theoretic language. It does **not** establish an unconditional critical-line theorem. The supplied papers explicitly retain an unproved global signed comparison. Their proved local results and operator identities cannot be substituted for that comparison.

Sources: [W] *weil_window(1).pdf*, especially Propositions 8.21, 9.15, 9.17, 9.30, 9.32 and 9.33; [T] *Michaels_Consolidated_Theta_Local_Amplitude_Proof 2.pdf*, opening status statement and CP20. References below use the printed numbering. The elementary lemmas proved here are independently justified; source-specific representation and boundary identities retain their stated hypotheses.

[T] reports strict positivity of its odd form on $0 < a \leq 3/8$, with margin greater than $1/200$. This is a local result. Identifying that form with another window formulation requires matching normalization, domain and parity. It does not by itself certify the proposed starting point $a_0 = 0.8$ in [W], or any larger support.

For the completion argument let Q_a be the relevant closed window form, $\lambda(a)$ its normalized variational minimum, and $A_0(a)$ the boundary coefficient of a normalized minimizer, wherever that coefficient is defined. The intended boundary identity is

$$\lambda'(a) = -2|A_0(a)|^2.$$

Its existence almost everywhere alone does not imply that it can be integrated to recover λ .

2. A rigorous cutoff for the prime adjacency

For $a > 0$, put

$$c_n = \frac{\Lambda(n)}{\sqrt{n}}, \quad B(a) = 2 \sum_{2 \leq n < e^{2a}} c_n.$$

Let C_a be the symmetric weighted adjacency of the confined echo graph, with steps $\pm \log n$ of weight c_n . This is the operator of [W, Proposition 8.21]. The vertices may be infinite in number, even though the set of allowed step sizes is finite.

Lemma 1 (explicit resolvent control). On the graph's square-summable vertex space,

$$\|C_a\| \leq B(a).$$

For any vector b_a , set $\beta(a) = \|b_a\|^2$. If $s > B(a)$, then

$$R_a(s) = \langle b_a, (sI - C_a)^{-1} b_a \rangle$$

satisfies

$$0 \leq R_a(s) \leq \frac{\beta(a)}{s - B(a)}, \quad \left| R_a(s) - \frac{\beta(a)}{s} \right| \leq \frac{\beta(a)B(a)}{s(s - B(a))}.$$

Consequently, if

$$S(a) = \max\{1, 2B(a), 2\sqrt{\beta(a)}\},$$

then for every $s \geq S(a)$,

$$s - R_a(s) \geq \frac{s}{2} > 0.$$

Proof. Each row and column has sum at most $B(a)$; the Schur bound gives the operator norm estimate, including for the infinite graph. The spectral theorem puts the spectrum in $[-B(a), B(a)]$, so the resolvent is positive and has norm at most $(s - B(a))^{-1}$. Moreover,

$$(sI - C_a)^{-1} - s^{-1}I = s^{-1}(sI - C_a)^{-1}C_a,$$

which proves the error bound. For $s \geq 2B(a)$, we have $R_a(s) \leq 2\beta(a)/s$. The condition $s \geq 2\sqrt{\beta(a)}$ gives the last inequality. $\square$

For the usual edge vector on one component, its direct prime-power entries give $\beta(a) = \sum_{n < e^{2a}} \Lambda(n)^2/n$, subject to the component convention of [W]. If endpoint components are combined, use their actual squared norm instead. Both this finite sum and $B(a)$ are locally bounded in a .

What this resolves. The abstract effective symbol $s - R_a(s)$ is positive beyond an explicit, locally bounded cutoff in its spectral parameter. This conclusion needs no hypothesis about zeta zeros and no asymptotic prime number theorem.

What it does not resolve. A cutoff for this resolvent is not a proved onset for the actual minimizer's Fourier asymptotic. In [W], $s = \tilde{\sigma}_\infty(t)$, asymptotically $\log(t/(2\pi)) - \lambda(a)$. Converting the cutoff to a uniform Fourier-tail statement requires quantitative control of that relation, a justified representation of the minimizer, and its remainder. [W, discussion after Proposition 8.21] explicitly distinguishes the dense-echo setting from the proved finite-echo representation. The lemma does not remove that distinction.

3. Why spectral mass and finite compression do not finish the comparison

The spectral measure μ_{b_a} of C_a obeys, for $u > 0$,

$$\mu_{b_a}([u, \infty)) \leq \frac{\|C_a b_a\|^2}{u^2}.$$

This follows directly by integrating $1_{[u, \infty)}(\nu) \leq \nu^2/u^2$. The sharper moment estimates proposed in [W, Proposition 9.33] concern this measure. Even a strong bound of this kind requires an additional transfer estimate before it bounds the Fourier tail of a normalized minimizer relative to $\lambda(a)$.

There is also a separate compression issue. Positivity on an edge/echo subspace does not imply positivity of the full form. Already

$$M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

has positive compressions to each coordinate, but $\langle (1, -1), M(1, -1) \rangle = -2$.

For bounded self-adjoint blocks, a precise sufficient replacement is the Schur complement: if

$$Q = \begin{pmatrix} D & E \\ E^* & G \end{pmatrix}, \quad G \geq \delta I > 0,$$

then $Q \geq 0$ exactly when $D - EG^{-1}E^* \geq 0$. This follows by completing the square in the second coordinate. For unbounded forms the corresponding domain and factorization hypotheses must first be supplied. Thus the compression identity in [W, Proposition 9.32] is useful arithmetic structure, but still needs control of its coupling to the complementary subspace.

4. A sufficient regularity lemma

Extend a window function by zero to the real line and define

$$g_f(d) = \int_{\mathbb{R}} f(x+d) \overline{f(x)} dx.$$

Lemma 2 (autocorrelation from bounded variation). If the zero extension of f is bounded and has finite total variation $V(f)$, then

$$|g_f(d+h) - g_f(d)| \leq \|f\|_{\infty} V(f) |h|$$

for every real d, h . In particular, every lower one-sided difference quotient of $\operatorname{Re} g_f$ is finite from below.

Proof. A function of bounded variation satisfies $\|f(\cdot+h) - f\|_1 \leq V(f)|h|$: express its increments by its finite distributional derivative measure and apply Fubini to its total variation. Multiplication by $\overline{f}$, followed by integration, gives the displayed estimate. The zero extension includes any endpoint jumps. $\square$

This provides a concrete sufficient condition for the autocorrelation finiteness hypothesis in [W, Proposition 9.17]. To use that proposition globally, one must establish its remaining hypotheses and establish this bounded-variation property, or another adequate finiteness condition, at every required support outside at most a countable exceptional set. A theorem only at almost every support does not supply that statement.

The distinction is substantive: let $\mathcal{C}$ be the Cantor function on $[0, 1]$. The continuous nonincreasing function $1 - \mathcal{C}$ has derivative zero almost everywhere but drops from one to zero. It therefore satisfies an almost-everywhere identity $\lambda' = -2|A|^2$ with $A = 0$, yet integrating that identity misses all its decrease. This example is about the logical regularity requirement, not a proposed zeta window.

5. The exact first-zero argument

Theorem 3 (conditional completion). Suppose $\lambda : [a_0, \infty) \rightarrow \mathbb{R}$ is locally absolutely continuous, $\lambda(a_0) > 0$, and

$$\lambda'(a) = -2|A_0(a)|^2$$

almost everywhere. Suppose there is a nonnegative function $K \in L^1_{\text{loc}}([a_0, \infty))$ such that, wherever $\lambda(a) > 0$,

$$2|A_0(a)|^2 \leq K(a)\lambda(a) \quad \text{almost everywhere.} \quad (*)$$

Then, for every finite $b \geq a_0$,

$$\lambda(b) \geq \lambda(a_0) \exp \left(- \int_{a_0}^b K(u) du \right) > 0.$$

Proof. If a first zero b_* existed, λ would be positive before it. On each compact subinterval preceding b_* , continuity gives a positive lower bound, so the absolutely continuous chain rule applies to $\log \lambda$. From (*) we obtain $(\log \lambda)' \geq -K$. Integrate up to $b < b_*$, then let $b \uparrow b_*$. Local integrability of K and continuity of λ contradict $\lambda(b_*) = 0$. The same integration gives the quantitative bound at every finite b . $\square$

Only local integrability of K is needed. Growth as $a \rightarrow \infty$, however large, is not itself an obstruction. Conversely, a coefficient locally integrable only on the unknown positive interval is insufficient: for $\lambda(a) = b_* - a$, the coefficient $K(a) = 1/(b_* - a)$ permits a finite first zero.

One may combine regularity and comparison into a measure inequality. If λ is continuous and nonincreasing, let $\nu = -d\lambda$ be its positive Stieltjes measure. The assertion

$$\nu(E) \leq \int_E K(a) \lambda(a) da$$

for every Borel set E on each compact positive interval, with K integrable across every prospective finite endpoint, excludes singular variation and yields the same first-zero argument. This is a sufficient target; the supplied almost-everywhere derivative identity does not prove it.

6. The remaining arithmetic statement

In the finite-tree regime, [W, Section 9] writes a minimizer as $f = P\phi_1$, with $P^2 = \|\phi_1\|_2^{-2}$, and selects the lowest spectral root by a secular equation. Its boundary amplitude satisfies $A_0 = PA_0^{(1)}$. Consequently (*) is exactly

$$2|A_0^{(1)}(a)|^2 \leq K(a)\lambda(a)\|\phi_1(a)\|_2^2$$

at that selected root, with an integrable coefficient on every finite support interval. This is the relative prime-side comparison still to be proved; outside the finite-tree regime an adequate full-window formulation is also necessary.

An absolute estimate $|A_0(a)|^2 \leq M(a)$ does not establish it. Division by $\lambda(a)$ gives $K = 2M/\lambda$, whose integral can diverge exactly at the first zero one is trying to exclude. The example in Section 5 makes that failure explicit.

An eventual Fourier-tail asymptotic also does not yield a locally uniform onset by itself. Its remainder must be controlled in the support parameter, including near possible spectral degeneracy and the exceptional supports. Lemma 1 supplies an operator cutoff, not this missing remainder estimate.

With the appropriate Weil-positivity equivalence and its test-space hypotheses established, Theorem 3 supplies the intended critical-line deduction. Neither that equivalence nor the deduction replaces the missing comparison. The source material supports a conditional completion with useful proved estimates, but does not supply an unconditional completion of all three requested steps. In a fictional arithmetic theory, (*) and the regularity requirement could be declared axioms; doing so would leave them premises, not derived number-theoretic theorems.

7. Dictionary

Earlier terminology	Number-theoretic meaning
Floor	Variational minimum $\lambda(a)$ of the window form
Edge amplitude	Boundary coefficient $A_0(a)$ of a normalized minimizer
Relative leak	$2 A_0(a) ^2/\lambda(a) = -\lambda'(a)/\lambda(a)$, when defined and justified
Horizon	Reference height $T^*(a) = 2\pi e^{2a}$
Echo graph	Confined graph of prime-power translation steps
Band	Spectrum of the prime adjacency, expressed in the relevant symbol parameter
Tail onset	Height after which a stated Fourier-tail estimate holds with controlled error
First null	First finite support with $\lambda(a) = 0$
Exceptional supports	Parameters where the available representation or regularity theorem does not apply

The dictionary does not identify an adjacency-band cutoff with a Fourier-tail onset. That additional identification is a mathematical assertion requiring proof.

8. Shared source representation for the margin and boundary coefficient

This continuation treats the odd sector. Write the operator of the form as

$$Q_a = H_a - v_a v_a^*, \quad v_a(x) = \sqrt{2} \sinh(x/2),$$

where H_a contains the archimedean contribution and all the signed prime translations. In particular, H_a is not the graph adjacency C_a . The following calculation assumes $H_a \geq h_a I > 0$. It does not assert that this hypothesis has already been proved at every support.

Fix a support and suppress its subscript. For $0 \leq z < h$, define

$$R_z = (H - zI)^{-1}, \quad u_z = R_z v, \quad m(z) = \langle v, R_z v \rangle, \quad \delta = 1 - m(0).$$

These are exact resolvents of the full operator, not finite-mode truncations.

Proposition 4 (polar decomposition). With $y = H^{1/2}f$, $w = H^{-1/2}v$, and $m = m(0) > 0$,

$$Q(f) = \left\| y - \frac{\langle w, y \rangle}{m} w \right\|^2 + \frac{1-m}{m} |\langle v, f \rangle|^2.$$

Hence $Q \geq 0$ if and only if $m \leq 1$. When $\delta > 0$,

$$Q(f) \geq \delta \langle f, Hf \rangle \geq h\delta \|f\|^2.$$

Proof. Orthogonally decompose y along w , whose squared norm is m . Subtract $|\langle w, y \rangle|^2$. Necessity of $m \leq 1$ follows by choosing $f = H^{-1}v$, for which $Q(f) = m(1-m)$. The quantitative inequality follows from $|\langle w, y \rangle|^2 \leq m\|y\|^2$. $\square$

Suppose now that $\lambda \in (0, h)$ is an eigenvalue of $H - vv^*$. Its eigenvector has nonzero overlap with v , since otherwise it would be an eigenvector of H below h . Consequently

$$m(\lambda) = 1, \quad f = \frac{u_\lambda}{\|u_\lambda\|}$$

up to phase. The resolvent identity gives two exact identities:

$$\boxed{u_\lambda = u_0 + \lambda R_\lambda u_0}, \quad \boxed{\delta = \lambda \langle u_0, u_\lambda \rangle}. \quad (8.1)$$

The inner product on the right is real and strictly positive.

Proposition 5 (quantitative equivalence of the margins). Put $N = \|u_\lambda\|^2$. Then

$$\delta \leq \lambda N \leq \frac{h}{h-\lambda} \delta, \quad \delta \geq \lambda \|u_0\|^2. \quad (8.2)$$

In particular, if $\lambda \leq h/2$, then

$$\delta \leq \lambda N \leq 2\delta.$$

Proof. Let μ be the spectral measure of H at v ; it is supported on $[h, \infty)$. Then

$$\delta = \lambda \int \frac{d\mu(t)}{t(t-\lambda)}, \quad \lambda N = \lambda \int \frac{d\mu(t)}{(t-\lambda)^2}.$$

The ratio of the second integrand to the first is $t/(t-\lambda)$, between one and $h/(h-\lambda)$. Finally $1/[t(t-\lambda)] \geq 1/t^2$. $\square$

Thus the proposed scalar margin is quantitatively comparable to the original eigenvalue denominator near zero, provided a positive lower bound for H is retained.

9. Transfer of the boundary coefficient to zero spectral parameter

Fix a phase convention for the boundary coefficient. Let $\mathcal{B}_a$ denote the resulting linear coefficient map on functions for which the same leading boundary profile is established. Its normalization is chosen so that its modulus agrees with the amplitude in the boundary derivative identity. Existence of this map on the following source solutions is an explicit hypothesis; square integrability alone does not give it.

At a fixed support assume $\mathcal{B}$ is defined on u_0 , u_λ , and $R_\lambda u_0$, using the same profile. Set

$$\alpha_0 = \mathcal{B}u_0, \quad \alpha_\lambda = \mathcal{B}u_\lambda, \quad \beta_\lambda = \mathcal{B}R_\lambda u_0.$$

Linearity and (8.1) prove

$$\boxed{\alpha_\lambda = \alpha_0 + \lambda\beta_\lambda.} \tag{9.1}$$

This is a common source representation of the two quantities: $\delta = 1 - \langle v, u_0 \rangle$, while α_0 is the boundary coefficient of the same zero-parameter source solution $u_0 = H^{-1}v$.

Proposition 6 (a sufficient source estimate). Suppose, over the support region in question,

$$|\alpha_0|^2 \leq M(a)\delta, \quad |\beta_\lambda| \leq D(a), \quad n(a) = \|u_0\|^2 > 0. \tag{9.2}$$

Then

$$2|\alpha_\lambda|^2 \leq \left(4M(a) + \frac{4D(a)^2\lambda(a)}{n(a)} \right) \delta. \tag{9.3}$$

Writing the parenthesis as $K(a)$, the normalized minimizer satisfies

$$2|A_0(a)|^2 \leq K(a)\lambda(a).$$

For the first-zero theorem, this coefficient must have an integrable majorant across every prospective finite endpoint.

Proof. Equation (9.1) and $|x+y|^2 \leq 2|x|^2 + 2|y|^2$ give $2|\alpha_\lambda|^2 \leq 4M\delta + 4\lambda^2 D^2$. By (8.2), $\lambda^2 \leq \lambda\delta/n$, proving (9.3). Divide by N and use $\delta \leq \lambda N$. $\square$

For example, local boundedness of M, D, λ together with a locally uniform positive lower bound for n would suffice for local boundedness of K . These properties have not been deduced here for the number-theoretic family.

At a prospective endpoint with $\delta \rightarrow 0$, a bounded M in (9.2) requires $\alpha_0 \rightarrow 0$, with the quantitative rate $|\alpha_0| = O(\sqrt{\delta})$. Vanishing at the endpoint alone is weaker than the rate and does not establish (9.2).

10. What the source formulas supply, and what still needs proof

The representation in [W, Corollary 7.24] retains a near-source term, rational corrections at real symbol zeros, and feedback from the opposite edge. Proposition 8.18 extends the transform representation using the full window symbol under its stated hypotheses. Its following discussion explicitly distinguishes that exact representation from the existence and uniformity of a limiting boundary amplitude.

In particular, the exterior feedback depends on the solution itself. Removing it would change the source equation. The representation does not provide an identity expressing $|\mathcal{B}H^{-1}v|^2$ as a

controlled multiple of $1 - \langle v, H^{-1}v \rangle$. Nor does the result stated for a critical point automatically give boundary estimates for both of the forced solutions $H^{-1}v$ and $(H - \lambda I)^{-1}H^{-1}v$. Those extensions require proof.

The arithmetic estimate isolated by this calculation is

$$\boxed{|\mathcal{B}_a H_a^{-1} v_a|^2 \leq M(a) (1 - \langle v_a, H_a^{-1} v_a \rangle)}. \quad (10.1)$$

Together with the controlled correction in (9.2), it would give the required relative comparison. Equation (10.1) remains a target, not a consequence of the resolvent identity.

There is an additional branch at a first null. If $Q \geq 0$, then $H = Q + vv^* \geq 0$, and

$$\ker H = \ker Q \cap v^\perp.$$

Indeed, $\langle f, Hf \rangle = Q(f) + |\langle v, f \rangle|^2$, a sum of nonnegative terms. If H has compact resolvent, the absence of this joint kernel makes H strictly positive at that support. A null vector orthogonal to v belongs to the joint kernel and must be treated separately. Uniform positivity near the support also needs the appropriate continuity argument.

Exact outcome of this continuation. The full-operator polar decomposition, source-transfer identities (8.1) and (9.1), and estimates (8.2) and (9.3) are proved under their explicit operator and boundary-domain hypotheses. They reduce the proposed comparison to quantitative control of the boundary coefficient of a zero-parameter forced solution, plus its resolvent correction. The supplied formulas have not established (10.1), the necessary source-domain bounds, or integration regularity at all required supports. No unconditional critical-line conclusion is claimed.

Additional dictionary entries

Symbol	Meaning in this continuation
H_a	Full odd window operator before subtracting the polar rank-one term; distinct from C_a
v_a	Polar source $\sqrt{2} \sinh(x/2)$ on the window
u_z	Forced solution $(H_a - zI)^{-1}v_a$
$m(z)$	Scalar resolvent value $\langle v_a, u_z \rangle$
δ_a	Scalar margin $1 - m(0)$; not an eigenvalue
$\mathcal{B}_a$	Boundary coefficient map on its justified domain
α_0	Coefficient of the zero-parameter forced solution, generally unnormalized
α_λ	Coefficient of the forced solution at the secular root
$A_0(a)$	Coefficient of the normalized minimizer; $ A_0 = \alpha_\lambda /\ u_\lambda\ $

11. The zero-parameter source has exactly the required small energy

Retain the hypotheses and notation of Sections 8-9, including the full signed prime operator. At a positive secular root $0 < \lambda < h$, write

$$u = H^{-1}v, \quad m = \langle v, u \rangle, \quad \delta = 1 - m.$$

Then $0 < m < 1$. The source equation gives the exact identities

$$\boxed{Qu = \delta v, \quad Q(u) = m\delta.} \quad (11.1)$$

Here the first Q denotes the associated operator and the second its quadratic form. Indeed, $Hu = v$, so $(H - vv^*)u = (1 - m)v$, and pairing with u gives $m(1 - m)$. No boundary estimate is used.

Thus the source at zero spectral parameter already has small full Weil energy as m approaches one. The problem is to control its boundary coefficient using that small energy, while retaining the signed low-frequency contribution and the polar subtraction.

Let Ψ_a be the real even Fourier symbol representing the nonpolar form on the window:

$$\Psi_a(t) = \operatorname{Re} \psi_\Gamma \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi - 2 \sum_{2 \leq n < e^{2a}} \frac{\Lambda(n)}{\sqrt{n}} \cos(t \log n),$$

where $\psi_\Gamma = \Gamma'/\Gamma$ is the digamma function. This is the source's normalization, with every active prime-power coefficient retained. For admissible g ,

$$\langle g, H_a g \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} \Psi_a(t) |\hat{g}(t)|^2 dt.$$

Choose T beyond the last sign change of $\Psi_a - \lambda$, so the relevant high-frequency integrands are nonnegative. Write $U = \hat{u}$ and set

$$E_{>}(u; T) = \frac{1}{2\pi} \int_{|t| > T} \Psi_a(t) |U(t)|^2 dt,$$

$$E_{\leq}(u; T) = \frac{1}{2\pi} \int_{|t| \leq T} \Psi_a(t) |U(t)|^2 dt - m^2.$$

All of the polar subtraction is included in $E_{\leq}$; this is a specified bookkeeping convention. By (11.1),

$$\boxed{E_{>}(u; T) + E_{\leq}(u; T) = m\delta.} \quad (11.2)$$

Proposition 7 (signed low-frequency estimate implies source comparison). Assume the boundary coefficient $\alpha_0 = \mathcal{B}_a u$ exists and the quantitative tail lower bound

$$E_{>}(u; T) \geq \frac{|\alpha_0|^2}{\pi T} \quad (11.3)$$

holds. If, for some $C(a) \geq 0$,

$$\boxed{E_{\leq}(u; T) \geq -C(a)m\delta,} \quad (11.4)$$

then

$$|\alpha_0|^2 \leq \pi T(1 + C(a))m\delta. \quad (11.5)$$

Proof. Equation (11.2) and (11.4) imply $E_{\geq} \leq (1 + C)m\delta$; combine this with (11.3). $\square$

This supplies the coefficient $M(a) = \pi T(1 + C(a))m$ in Proposition 6. The arithmetic obligation (11.4) has no boundary coefficient in its statement. It asks that the low-frequency part, including the complete polar subtraction, not be too negative compared with the full source energy. It remains unproved. Positivity of Ψ_a at high frequencies alone does not imply it.

12. Quantitative control of the resolvent correction from a tail bound

The correction source is

$$w = (H_a - \lambda I)^{-1}u, \quad (H_a - \lambda I)w = u, \quad \beta_\lambda = \mathcal{B}_a w.$$

Define

$$J(a, \lambda, T) = \max_{|t| \leq T} |\Psi_a(t) - \lambda|, \quad V(a) = \|v_a\|.$$

The explicit source norm is $V(a)^2 = 2(\sinh a - a)$.

Proposition 8 (forced-source boundary bound). Assume the boundary coefficient of w exists, and its tail obeys

$$\frac{1}{2\pi} \int_{|t| > T} (\Psi_a(t) - \lambda) |\widehat{w}(t)|^2 dt \geq \frac{|\beta_\lambda|^2}{\pi T}. \quad (12.1)$$

Then

$$\boxed{|\beta_\lambda|^2 \leq D(a)^2 := \frac{\pi T V(a)^2}{h^2} \left(\frac{1}{h - \lambda} + \frac{J(a, \lambda, T)}{(h - \lambda)^2} \right)}. \quad (12.2)$$

Proof. Pairing the forced equation with w and using the form representation gives

$$\frac{1}{2\pi} \int_{\mathbb{R}} (\Psi_a - \lambda) |\widehat{w}|^2 = \langle w, u \rangle.$$

The integral over $|t| \leq T$ is at least $-J\|w\|^2$, by Parseval. Therefore the tail is at most $\|w\|\|u\| + J\|w\|^2$. The resolvent bounds

$$\|u\| \leq \frac{V}{h}, \quad \|w\| \leq \frac{V}{h(h - \lambda)}$$

and (12.1) prove (12.2). $\square$

This is a forced-equation extension of the energy splitting used in [W, Lemma 9.14]. Its tail hypothesis is stated explicitly: the lemma for a minimizing eigenfunction cannot simply be applied to w , which solves a different equation.

Combining Propositions 6-8 yields the fully specified coefficient

$$K(a) = 4\pi T(1 + C(a))m + \frac{4\lambda(a)D(a)^2}{\|H_a^{-1}v_a\|^2} \quad (12.3)$$

in $2|A_0(a)|^2 \leq K(a)\lambda(a)$. The same cutoff T may be used when both tail lower bounds hold beyond their respective onsets, by choosing the larger threshold.

For a prospective first zero, suppose $H_a \geq h_0 I > 0$ uniformly nearby and $\lambda(a) \leq h_0/2$. If T, C, J, V are locally bounded there and $m \geq 1/2$, then K is locally bounded: Cauchy-Schwarz gives

$$\|H_a^{-1}v_a\|^2 \geq \frac{m^2}{V^2} \geq \frac{1}{4V^2},$$

and (12.2) has denominators bounded away from zero. These are sufficient conditions, not established properties of the full number-theoretic family.

Outcome. The correction bound now follows from a stated forced-source tail bound and a positive lower bound for H_a . The small-energy source bound follows from a signed low-frequency estimate. What remains is to prove (11.4), the two locally controlled tail-onset statements for these actual forced solutions, and the regularity needed for the first-zero integration. The calculation has linked the source formulation to the existing three obligations without assuming that any of them is already solved.